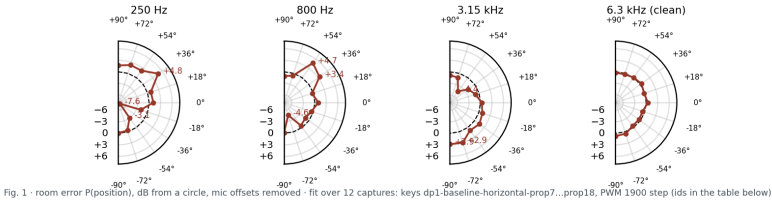


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2026-09-08 · rev 2
data: laptop backup of 2 Sept
scripts: arc_error_map.py, arc_validation_diagnostics.py
full guide: docs/arc-validation-guide.html
remedies report: docs/arc-validation-remedies.pdf

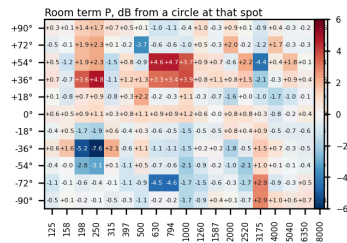
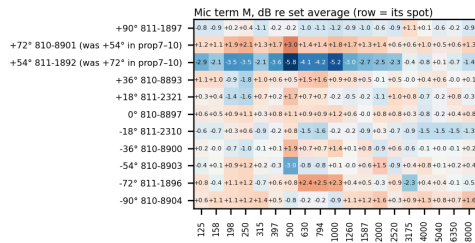
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How to read the map. Row = spot on the arc, column = frequency band. White: the spot reads what a circle predicts. Red: too loud. Blue: too quiet. Each microphone's own error is already taken out (left panel), so the colour is the room.

The 200–250 Hz column is the blade tone, and it is not one nearby surface. At BPF 216 Hz the tone polar is round to ± 2 dB; at 238 Hz -36° reads -8 dB; at 258 Hz the hole moves to -72° and the peak to $+54^\circ$, while the broadband floor beside the tone stays round to ± 1 dB. A surface within a metre cannot change the pattern that fast; narrow room modes sampled by a line spectrum can, and so can a propeller whose inflow is not axisymmetric. A loudspeaker at the hub decides which (remedies report, \$03, \$10).

White above 4 kHz: every spot reads the same. Two reasons. The foam absorbs short waves well (under 9 cm above 4 kHz), so little is reflected. And a band at 4 kHz is 900 Hz wide, so a reflection cancels and adds many times inside it and averages out; at 250 Hz the band is 58 Hz wide and holds one cancellation, so it shows. White means the arc and the microphones work; the problem is limited to the low bands.



Three suspects, one per band in Fig. 2. Each is a size and a place, read off the map: a surface shows up once the wavelength is shorter than it (Rajmane and Baumann, DAGA 2016: a 35 cm plate from 1.25 kHz, a 10 cm plate from 3.15 kHz).

1. Put 811-1892 off the arc.
2. Put a small loudspeaker on the stand at the hub. Play a 20 Hz–20 kHz sine sweep and record all microphones; each microphone's own impulse response gives its echo delays (no clock sync needed), times 343 m/s, the extra paths. Then play tones at 216, 238 and 258 Hz: if the ± 8 dB pattern of Fig. 1 comes back, it is the room; if not, it is the propeller's inflow.
3. Write down the geometry in docs/mic_arc.md: arc radius, propeller height, which wall or corner is at the -90° end, distances from the -72° and $+72^\circ$ microphones to the nearest wall.
4. Run a ladder of 12–15 motor speeds (BPF every ~ 5 Hz, 205–265 Hz) and average the tone polar over it; cover or move the surfaces the sweep names; rerun the fit. Foam works from 650 Hz up; at 250 Hz the levers are averaging, correction and distance, then resonant absorbers.

Every level in the twelve runs (11 microphones $\times$ 19 third-octave bands $\times$ 12 runs) is written as three parts added together: **run gain** + **position error** + **microphone offset**. The three parts are found by least squares, i.e. the single set of values that fits all 12×11 levels best in every band. **Run gain** takes up any level difference between runs (RPM, day, load). **Microphone offset** takes up each serial's own error after calibration. **Position error** is what is left that belongs to a spot on the arc; it is forced to average zero over the eleven spots, so it reads as dB from a circle. That is what Figs. 1 and 2 show.

Refitting from the data, each reproduces the maps to 0.01 dB; each map sums to zero per band by construction. 132 cells per band are explained by 34 unknowns; 87% of them are within 2 dB of gain + P + M. Scatter is 0.6–0.8 dB in the quiet bands and 1.5–2.3 dB in 200–250 Hz and 500–1000 Hz, the bands where the room acts. The worst cells (+11 dB at $-36^{\circ}/250$ Hz in prop7; +6 to +7 dB at $0^{\circ}/2-4$ kHz in prop7–8) come from prop7, which spun at 164 Hz where every other run spun at 238 Hz: a different speed; leaving it out moves the maps by under 1 dB. Limit: the 0° microphone (810-8897) never moved, so its M and the 0° P are known only as a sum.

+54° at 800 Hz: prop8 with 810-8901 reads +5.6 re gain = P +4.7 + M +1.4; prop18 with 811-1892 reads +3.1 = +4.7 - 4.2. Two mics, one room value.